

REQUIREMENT ANALYSIS

IntelliDocs AI — Intelligent Document Processing & RAG-Based Knowledge Platform

Document Type: Software Requirement Analysis

Technology Stack: ASP.NET Core, SQL Server, React.js, RAG / LLM

Version 1.0

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Introduction

- IntelliDocs AI is an AI-powered document intelligence platform designed to automate document processing, information extraction, document classification, summarization, semantic search, and question answering.
 - The system allows users to upload different types of documents such as PDFs, invoices, contracts, reports, resumes, certificates, and scanned documents. Using OCR, Natural Language Processing (NLP), embeddings, vector search, and Large Language Models (LLMs), the system converts unstructured documents into useful and searchable information.
 - The backend will be developed using ASP.NET Core Web API and C#, with SQL Server for structured data storage and React.js for the frontend.
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Problem Statement

- Organizations and individuals handle large volumes of documents every day. Manually reading, organizing, searching, and extracting information from these documents is time-consuming and error-prone.
 - Traditional document management systems mainly provide file storage and keyword-based search. They generally cannot understand the context or meaning of the information contained within documents.
 - For example, finding the answer to "What is the termination period mentioned in this contract?" may require manually searching through several pages.
 - IntelliDocs AI aims to solve this problem by providing an intelligent platform that can understand documents and allow users to interact with them using natural language.
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Objectives

The main objectives of IntelliDocs AI are:

- To provide a centralized platform for uploading and managing documents.
- To automatically extract text from digital and scanned documents.
- To classify documents based on their content.
- To extract important information such as names, dates, amounts, organizations, and document numbers.
- To generate concise AI-powered summaries.
- To provide semantic search across uploaded documents.
- To implement RAG-based question answering.
- To allow users to ask questions about one or multiple documents.
- To provide source references for AI-generated answers.

- To securely manage users and their documents.
 - To reduce the time required for manual document analysis.
 - To provide a scalable architecture using .NET and modern AI technologies.
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Scope of the System

In Scope

The system will support:

- User registration and login
- Secure document upload
- PDF and image-based document processing
- OCR-based text extraction
- Document classification
- Metadata extraction
- AI-powered summarization
- Document indexing
- Semantic search
- Vector-based retrieval
- RAG-based question answering
- Multi-document conversations
- Document comparison
- Chat history
- Document management
- Role-based access
- Source/citation display
- Dashboard and analytics

Out of Scope

The initial version will not include:

- Fully automated legal or financial decision-making
- Editing the original uploaded documents
- Training a new Large Language Model from scratch
- Guaranteed legal, medical, or financial advice
- Offline AI processing for very large models
- Automatic modification of official documents

Stakeholders

The primary stakeholders are:

End Users

Users who upload and analyze documents. They can:

- Upload documents
- Search documents
- Ask questions
- Generate summaries
- Compare documents
- View extracted information

Administrator

The administrator manages the platform and monitors system activity. Responsibilities include:

- Managing users
- Monitoring documents
- Viewing system statistics
- Managing access permissions
- Monitoring system errors

System Administrator / Developer

Responsible for:

- System deployment
- Database management
- API maintenance
- AI service configuration
- Security
- Monitoring and maintenance

Functional Requirements

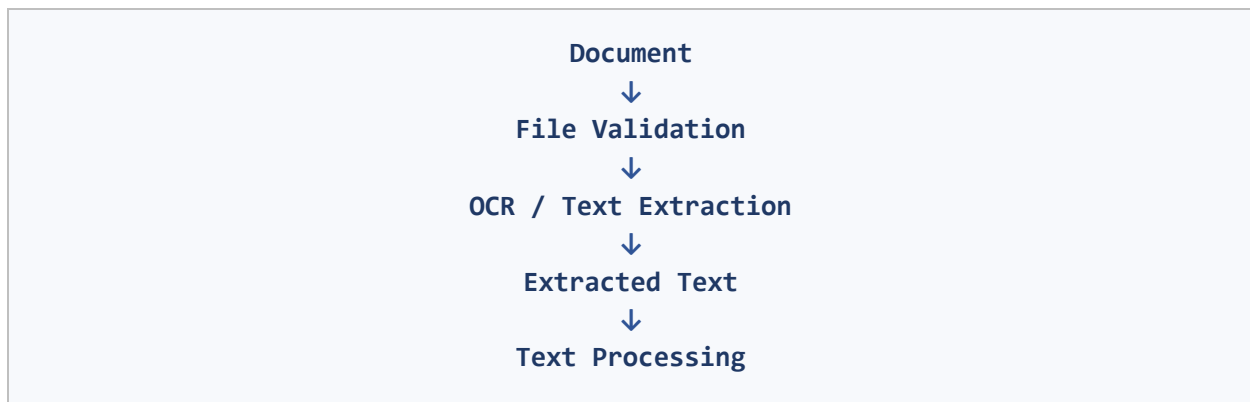
The following table summarizes the functional requirements of IntelliDocs AI:

ID	Requirement	Description
FR-01	User Registration	Allows new users to create an account using name, email, and password, with validation before the account is created.
FR-02	User Authentication	Allows registered users to securely log in using JWT-based authentication, with credential validation and session management.
FR-03	Document Upload	Allows authenticated users to upload PDF, PNG, JPG/JPEG, and DOCX files, with validation of file type, size, and integrity.
FR-04	Document Storage	Securely stores uploaded documents along with metadata such as document ID, owner, file type, size, and processing status.
FR-05	OCR and Text Extraction	Extracts text from uploaded documents, using OCR to convert scanned images into machine-readable text.
FR-06	Document Classification	Automatically classifies documents into categories such as invoice, contract, resume, report, or legal document.
FR-07	Information Extraction	Identifies key entities such as names, dates, amounts, and document numbers, and displays them to the user.
FR-08	AI Document Summarization	Generates short or detailed summaries, key points, and important entities from uploaded documents, processing large files in chunks.
FR-09	Document Chunking	Divides extracted text into smaller logical chunks to improve retrieval accuracy before embedding generation.
FR-10	Embedding Generation	Converts document chunks into vector embeddings representing their semantic meaning, stored in a vector-search system.
FR-11	Semantic Search	Allows users to search documents using natural language, retrieving results based on meaning rather than exact keyword matches.
FR-12	RAG-Based Question Answering	Provides a conversational interface that retrieves relevant document chunks and generates answers using an LLM.
FR-13	Source References	Displays the source document, page, and section supporting each AI-generated answer to improve transparency.
FR-14	Multi-Document Question Answering	Allows users to ask questions spanning multiple documents and receive a comparative, consolidated response.
FR-15	Document Comparison	Compares two documents and identifies added, removed, or modified content, including changed dates and amounts.
FR-16	Chat History	Stores previous document-related conversations; users can view, continue, or delete them.
FR-17	Document Management	Allows users to view, search, filter, download, and delete documents, and view metadata and processing status.

ID	Requirement	Description
FR-18	Dashboard	Displays document counts, processing status, category breakdown, storage usage, and AI query statistics.
FR-19	Role-Based Access Control	Defines separate permissions for regular users (personal documents and AI features) and administrators (user and system management).
FR-20	Error Handling	Handles upload, OCR, AI service, and database errors, and returns meaningful error messages to the user.

OCR and RAG Processing Flow

FR-05 (OCR and Text Extraction) and FR-12 (RAG-Based Question Answering) follow the processing flows below:



Non-Functional Requirements

Performance

The system should:

- Process normal documents within an acceptable time.
- Return search results quickly.
- Minimize unnecessary AI API calls.
- Support multiple concurrent users.

Security

Since documents may contain sensitive information, security is a major requirement. The system shall implement:

- JWT authentication
- Password hashing
- Role-based authorization
- Secure file access
- Input validation
- API authorization
- HTTPS
- Protection against unauthorized document access

Users must only be able to access documents for which they have permission.

Reliability

The system should:

- Handle failed AI requests gracefully.
- Prevent data loss during document processing.
- Maintain consistent database records.
- Provide appropriate error logging.

Scalability

The architecture should support increasing:

- Number of users
- Number of documents
- Document size
- AI requests
- Database records

The backend should use a modular architecture so AI services can be replaced or upgraded independently.

Usability

The user interface should be:

- Simple
- Responsive
- Easy to navigate
- Accessible on desktop and mobile
- Clear about document-processing status

Maintainability

The system shall follow clean software engineering practices including:

- Layered architecture
- Separation of concerns
- Dependency injection
- Repository / service patterns where appropriate
- RESTful API design
- Centralized error handling
- Logging
- Environment-based configuration

Technology Requirements

Component	Technology
Frontend	React.js
Styling	Tailwind CSS
Backend	ASP.NET Core Web API
Programming Language	C#
ORM	Entity Framework Core
Database	Microsoft SQL Server
Authentication	ASP.NET Core Identity + JWT
OCR	Azure AI Document Intelligence

Component	Technology
AI / LLM	Azure OpenAI / compatible LLM
Embeddings	Embedding model
Vector Search	Azure AI Search / Vector Database
API Communication	REST
Version Control	Git + GitHub
Deployment	Microsoft Azure

Hardware Requirements

Minimum

- Processor: Intel Core i5 or equivalent
- RAM: 8 GB
- Storage: 10 GB available
- Stable internet connection
- Modern web browser

Recommended

- Processor: Intel Core i5/i7 or equivalent
- RAM: 16 GB
- SSD storage
- Stable broadband connection

Since the major AI processing can be performed using cloud services, a dedicated high-end GPU is not mandatory.

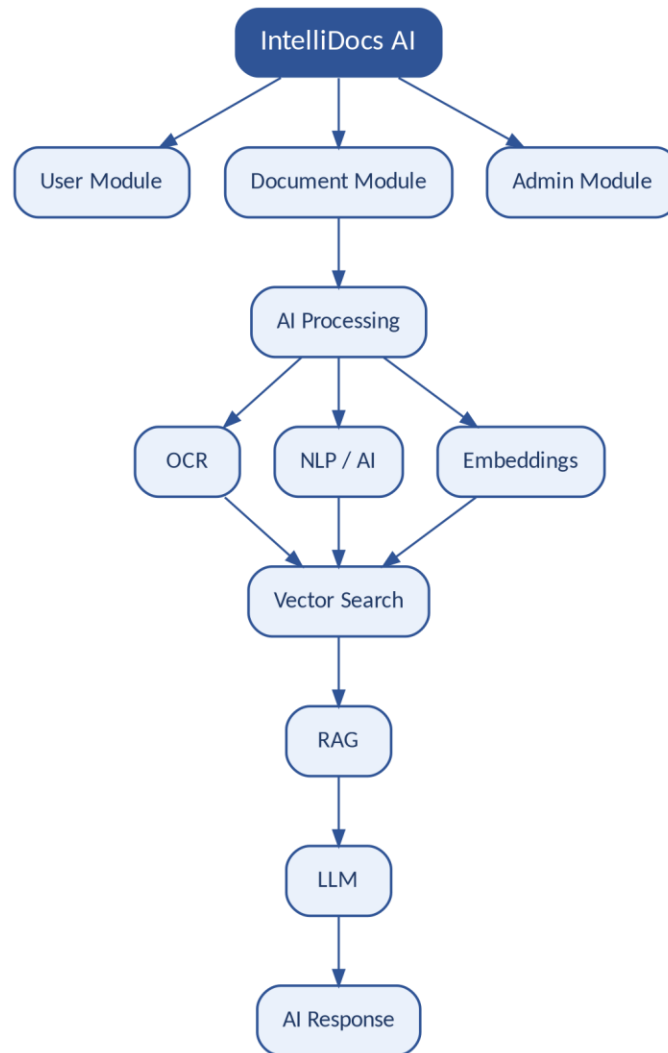
Software Requirements

- Windows / Linux / macOS
- .NET SDK
- Visual Studio / Visual Studio Code
- Node.js and npm

- React.js
 - SQL Server
 - SQL Server Management Studio
 - Git
 - GitHub
 - Azure account
 - Modern web browser
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High-Level System Requirements

The complete system can be divided into the following modules:



Expected Output

After successful implementation, IntelliDocs AI should provide a platform where a user can:

1. Create an account.
2. Upload documents.
3. Automatically extract text.
4. Identify the document type.
5. Extract important information.

6. Generate summaries.
 7. Search documents semantically.
 8. Ask questions using natural language.
 9. Receive answers based on document content.
 10. View sources supporting the answers.
 11. Compare multiple documents.
 12. Manage documents and conversation history securely.
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Future Enhancements

Future versions can include:

- Voice-based document queries
 - Multilingual document processing
 - Handwriting recognition
 - Automatic email/document ingestion
 - Advanced fraud/anomaly detection
 - Document version tracking
 - AI-generated reports
 - Team-based document sharing
 - Fine-tuned domain-specific models
 - Mobile application
 - Advanced analytics
 - Automated workflow generation
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Conclusion

IntelliDocs AI aims to transform traditional document management into an intelligent knowledge platform. By combining ASP.NET Core, SQL Server, React.js, OCR, NLP, embeddings, vector search, and RAG-based LLM technology, the system can convert unstructured documents into searchable and actionable knowledge.

The project provides a practical implementation of both modern .NET software engineering and artificial intelligence, making it suitable for demonstrating skills in full-stack development, cloud technologies, database management, AI integration, and system architecture.